

Radial Completion of Factor-Ray Dynamics: A GCD–Mellin Joint Spectrum, Exact Anisotropic Dephasing, and the Fixed-Shift Boundary

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

Put $u(n) = \Lambda(n) - 1$, let $W \in C_c^1((0, \infty); \mathbb{R})$ with $\text{supp } W \subset [\alpha, \beta] \subset (0, \infty)$, and consider the two-coordinate residual

$$\mathcal{R}_{h,D}^{\text{rad}}(X; t, s) = \sum_{\substack{n > D \\ m \geq 1}} u(n)(\Lambda(mn + h) - 1)W(mn/X)(n/m)^{it}(m, n)^{is}.$$

Writing $n = ak$, $m = bk$, $(a, b) = 1$, turns its joint frequency into $(\log(a/b), \log k)$. Unlike the previous ratio-only Mellin signal, this GCD–Mellin lift separates every factor pair. We formulate the lift as a commuting two-parameter unitary action and prove exact coefficient recovery, a tensor Fejér Parseval identity, and a rectangular Montgomery–Vaughan estimate. The lift is a ray-adapted coordinate system, not an information-theoretic enlargement of the full ratio–product two-Mellin continuum used in TPC-10.

On a sector $R \leq ab < 2R$, the two frequency gaps are bounded below by

$$\delta_{\text{ray}} = 2 \operatorname{arsinh}(1/(4R)), \quad \delta_{\text{rad}} = \log\left(1 + \sqrt{R/(\beta X)}\right).$$

Consequently exact tensor-Fejér dephasing occurs at the separate scale parameters $T_{\text{ray}} \asymp R$ and $T_{\text{rad}} \asymp \sqrt{X/R}$. Both scales are sharp up to constants along populated interior sector families; their sectorwise invariant is $T_{\text{ray}}T_{\text{rad}}^2 \asymp X$. This is not a single global bandwidth law: the undecomposed signal has the worse simultaneous scales X and $\sqrt{X}$. Hard rectangles are asymptotic only when both side lengths exceed the corresponding reciprocal gaps by unbounded factors.

The fully dephased energy is exactly

$$\mathcal{F}_{h,D}(X) = \sum_{\substack{n > D \\ m \geq 1}} u(n)^2(\Lambda(mn + h) - 1)^2 |W(mn/X)|^2.$$

Let $G \in C_c^1((0, \infty))$ be nonnegative and nonzero, and put $I_0(G) = \int_0^\infty G(v) dv$ and $I_2(W) = \int_0^\infty |W(v)|^2 dv$. Using the weighted exceptional-set transfer from TPC-12, for fixed $0 < \varepsilon < 13/15$ and $0 < \eta < 1$, uniformly in $X^{2/15+\varepsilon} \leq H \leq X$ and $1 \leq D \leq X^{1-\eta}$, we prove

$$\begin{aligned} \sum_h G(h/H) \mathcal{F}_{h,D}(X) &= \frac{1}{2} H X I_0(G) I_2(W) \log X ((\log X)^2 - (\log D)^2) \\ &\quad + O(H X (\log X)^2). \end{aligned}$$

The same result holds for the raw target. Radial dephasing removes the same-ray off-diagonal and hence the $\log \log X$ loss in TPC-12. The unnormalized parity-centered sum over even shifts has coefficient $1/4$.

The added coordinate repairs geometric information loss, not the fixed-shift arithmetic. We prove that the $m = 1$ endpoint of the dephased energy is, up to $O(X \log X)$, $(\log X)^2$ times the weighted Hardy–Littlewood prime-pair correlation. Thus its predicted fixed-even-shift asymptotic is equivalent to that open correlation; for $h = 2$ it is the weighted Hardy–Littlewood twin-prime asymptotic, which is strictly stronger than mere infinitude of twin primes. No fixed-shift prime-pair asymptotic, twin-prime lower bound, new level of distribution, or breach of sieve parity is claimed.

Keywords: factor-ray dynamics; GCD–Mellin spectrum; anisotropic dephasing; primes in almost all short intervals; prime-pair boundary; sieve parity.

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1 Introduction and scope

The factor-ray program has reached a precise coordinate bottleneck. The ratio phase $(n/m)^{it}$ distinguishes primitive rays but is constant along each ray. If

$$n = ak, \quad m = bk, \quad (a, b) = 1,$$

then $(n/m)^{it} = (a/b)^{it}$ and the radial coordinate $k = (m, n)$ disappears. TPC-10 had already used the full ratio–product two-Mellin continuum for endpoint selection [9]; that pair of variables is information-theoretically complete on a finite factor box. TPC-11 then isolated the ratio coordinate, proved exact long-band dephasing of the distinct ratios, and identified the remaining same-ray arithmetic [10]. TPC-12 evaluated the squared ray coefficients after averaging the shift over

$$X^{2/15+\varepsilon} \leq H \leq X$$

by transferring the almost-all short-interval theorem of Guth and Maynard [3] through an unbounded divisor weight [11]. Its fixed-ray minimax theorem also showed that no amount of ratio bandwidth can reconstruct the lost k -profile.

The next step should therefore restore the missing coordinate directly, not formally set $H = 1$. Relative to the ratio-only signal, we observe

$$(m, n)^{is} = k^{is}. \tag{1}$$

It is important to distinguish this phase from a second product Mellin phase such as $(mn/X)^{is}$. Since $mn = abk^2$, the latter mixes the primitive-ray product ab with k ; it is not transformed into (1) by a global linear change of the two old frequency variables. After a primitive ray is known, however, a ray-dependent shift and rescaling recovers $\log k$ from the product frequency. Thus the GCD–Mellin coordinate is a new directly observed, ray-adapted parametrization, but it is not richer than TPC-10’s full ratio–product continuum.

The classical Mellin transform is Fourier analysis on the multiplicative group [1]. Our product-Fejér identities are elementary tensor Fourier inversion, and the hard-rectangle estimate is obtained by applying the one-dimensional Montgomery–Vaughan inequality twice. We do not claim a new general multidimensional Mellin, Plancherel, or large-sieve theorem. The novelty is the direct GCD lift of this factor-ray coefficient system, its explicit anisotropic sector gaps, and the arithmetic transfer for the resulting diagonal observable.

1.1 The two-coordinate lift

Throughout,

$$u(n) = \Lambda(n) - 1, \quad \eta(q) = \Lambda(q) - 1,$$

and $W \in C_c^1((0, \infty); \mathbb{R})$ has support in a fixed interval $[\alpha, \beta] \subset (0, \infty)$. We extend $\Lambda(q) = 0$ for nonpositive integers q ; in every asymptotic application X is large enough that the supported targets are positive. Define

$$\mathcal{T}_{h,D}^{\text{rad}}(X; t, s) := \sum_{\substack{n > D \\ m \geq 1}} u(n) \Lambda(mn + h) W(mn/X) (n/m)^{it} (m, n)^{is}, \quad (2)$$

$$\mathcal{M}_D^{\text{rad}}(X; t, s) := \sum_{\substack{n > D \\ m \geq 1}} u(n) W(mn/X) (n/m)^{it} (m, n)^{is}, \quad (3)$$

$$\mathcal{R}_{h,D}^{\text{rad}} := \mathcal{T}_{h,D}^{\text{rad}} - \mathcal{M}_D^{\text{rad}}. \quad (4)$$

The unique GCD decomposition gives

$$\mathcal{R}_{h,D}^{\text{rad}}(X; t, s) = \sum_{\substack{a, b \geq 1 \\ (a, b) = 1}} \sum_{\substack{k \geq 1 \\ ak > D}} x_{a,b,k}(h; X, D) (a/b)^{it} k^{is}, \quad (5)$$

where

$$x_{a,b,k}(h; X, D) := u(ak) \eta(abk^2 + h) W(abk^2/X). \quad (6)$$

The frequency map

$$(a, b, k) \mapsto (\log(a/b), \log k)$$

is injective. In dynamical language, it is the joint spectrum of two commuting diagonal generators. This is a finite-support unitary representation attached to the factor coordinates; it is not a claim that prime gaps or zeta zeros are eigenvalues of a physical or transfer operator.

The corresponding fully separated energy is

$$\mathcal{F}_{h,D}(X) := \sum_{\substack{a, b \geq 1 \\ (a, b) = 1}} \sum_{\substack{k \geq 1 \\ ak > D}} |x_{a,b,k}(h; X, D)|^2 \quad (7)$$

$$= \sum_{\substack{n > D \\ m \geq 1}} u(n)^2 \eta(mn + h)^2 |W(mn/X)|^2. \quad (8)$$

The second line follows from the same bijection. It is the target-square diagonal before any radial indices are combined.

1.2 Main advances

There are three distinct conclusions.

First, on the dyadic factor-ray sector $R \leq ab < 2R$, primitive ratios are separated at scale R^{-1} and radial logarithms at scale $\sqrt{R/X}$. A tensor Fejér kernel therefore recovers every coefficient and its squared norm exactly once the two scale parameters separately satisfy

$$T_{\text{ray}} \gg R, \quad T_{\text{rad}} \gg \sqrt{X/R}.$$

Both inequalities are sharp up to constants along populated interior sector families with adjacent rays or consecutive radial indices. The relation

$$T_{\text{ray}} T_{\text{rad}}^2 \asymp X$$

is consequently a sectorwise resolution identity, not permission to trade one missing coordinate bandwidth against the other. For hard rectangles, an asymptotic requires both side lengths to exceed the displayed reciprocal-gap scales by unbounded factors.

Second, the arithmetic average of (7) has a smaller error than the ratio-only ray energy. For nonnegative, nonzero $G \in C_c^1((0, \infty))$, fixed $0 < \varepsilon < 13/15$ and $0 < \eta < 1$, uniformly for

$$X^{2/15+\varepsilon} \leq H \leq X, \quad 1 \leq D \leq X^{1-\eta}, \quad (9)$$

we prove

$$\begin{aligned} \sum_h G(h/H) \mathcal{F}_{h,D}(X) &= \frac{1}{2} H X I_0(G) I_2(W) \log X \\ &\quad \times ((\log X)^2 - (\log D)^2) + O(H X (\log X)^2), \end{aligned} \quad (10)$$

where

$$I_0(G) = \int_0^\infty G(v) dv, \quad I_2(W) = \int_0^\infty |W(v)|^2 dv.$$

The raw target has the same formula. The $\log \log X$ loss in TPC-12 came from bounding $k \neq \ell$ on one ray by a two-target upper sieve. Here those terms are geometrically absent. A parity-centered average over even shifts has coefficient $1/4$ and the same error order.

Third, exact reconstruction does not evaluate the recovered arithmetic data. The endpoint slice $m = 1$ of (8) satisfies, for each fixed nonzero h ,

$$\mathcal{F}_h^{\text{end}}(X) = (\log X)^2 \sum_n \Lambda(n) \Lambda(n+h) |W(n/X)|^2 + O_{h,W}(X \log X). \quad (11)$$

Thus the predicted fixed-even-shift asymptotic for this positive spectral slice is equivalent to the corresponding weighted Hardy–Littlewood prime-pair asymptotic. The new coordinate reaches the boundary cleanly; it does not cross it.

1.3 Claim ledger

Statement	Status here	What it does not imply
Joint GCD–Mellin spectrum	Exact and injective on factor pairs	Not recovered from ratio-only data
Sector dephasing and reconstruction	Exact after stated filtering	No arithmetic coefficient is estimated
Short-shift radial energy	Unconditional in (9)	No prescribed shift is selected
Even-shift ensemble	Unconditional after parity centering	No conclusion for $h = 2$
Fixed endpoint	Exact reduction (11)	Its predicted asymptotic remains open

No fixed-shift Hardy–Littlewood asymptotic, twin-prime lower bound, prime number theorem for a fixed quadratic polynomial, new level of distribution, or violation of the parity barrier is asserted.

2 The joint factor-coordinate dynamics

We first isolate the exact algebra, independently of the prime weights.

Proposition 2.1 (Primitive-ray and radial coordinates). *The map*

$$(m, n) \mapsto \left(a = \frac{n}{(m, n)}, \quad b = \frac{m}{(m, n)}, \quad k = (m, n) \right) \quad (12)$$

is a bijection from $\mathbb{N}^2$ to the triples

$$(a, b, k) \in \mathbb{N}^3, \quad (a, b) = 1.$$

Moreover,

$$mn = abk^2, \quad \frac{n}{m} = \frac{a}{b}, \quad (m, n) = k. \quad (13)$$

Consequently the joint frequency map

$$(a, b, k) \mapsto \lambda_{a,b,k} := (\log(a/b), \log k) \quad (14)$$

is injective.

Proof. The GCD decomposition gives positive coprime a, b and recovers $n = ak$, $m = bk$, proving bijectivity and (13). If two joint frequencies agree, then $a/b = c/d$. Reduced positive fractions are unique, so $(a, b) = (c, d)$; equality of $\log k$ and $\log \ell$ then gives $k = \ell$. $\square$

The ratio-only map is the projection of (14) onto its first coordinate. Its fibre over a primitive ray (a, b) is the entire supported set of radial integers k . This is exactly the compression diagnosed in TPC-12.

2.1 A commuting unitary representation

Let $\mathcal{J} \subset \{(a, b, k) : (a, b) = 1\}$ be any finite index set and put $\mathcal{H}_{\mathcal{J}} = \ell^2(\mathcal{J})$. On its standard basis define two diagonal self-adjoint operators

$$L_{\text{ray}} e_{a,b,k} = \log(a/b) e_{a,b,k}, \quad L_{\text{rad}} e_{a,b,k} = \log k e_{a,b,k}. \quad (15)$$

They commute, and hence generate the unitary $\mathbb{R}^2$ -action

$$U(t, s) = \exp(itL_{\text{ray}} + isL_{\text{rad}}). \quad (16)$$

For a coefficient vector $x = (x_{a,b,k})$, the coordinate-sum observation is

$$\mathcal{O}_x(t, s) := \sum_{(a,b,k) \in \mathcal{J}} x_{a,b,k} e^{it \log(a/b) + is \log k}. \quad (17)$$

The arithmetic transform (5) is this observation with the finite set and coefficient vector selected by X, h, D, W .

Proposition 2.2 (Projection loss and joint completeness). *For a fixed primitive ray (a, b) , the restriction $s = 0$ observes only*

$$e^{it \log(a/b)} \sum_k x_{a,b,k}. \quad (18)$$

In contrast, the full function $(t, s) \mapsto \mathcal{O}_x(t, s)$ uniquely determines every coefficient $x_{a,b,k}$.

Proof. The first assertion is immediate. For the second, the exponentials have distinct frequency vectors by proposition 2.1. A finite linear combination of distinct characters of $\mathbb{R}^2$ cannot vanish identically. Explicit finite-resolution, Fejér-weighted recovery with quantitative sector thresholds is proved in theorem 4.2. $\square$

Remark 2.3 (What “dynamics” means here). The action (16) is an exact diagonal unitary action on the augmented factor-coordinate coefficient space. Its generators and joint eigenvalues are explicitly defined logarithmic coordinates. They are not asserted to be Koopman or transfer-operator eigenvalues of the logistic map, and they have no claimed identification with Riemann-zero ordinates. This separation of meanings is essential to the scope of the paper.

Remark 2.4 (Why a product phase is not the same lift). Adding $(mn/X)^{is}$ would attach the frequency $\log(abk^2/X) = \log(ab/X) + 2 \log k$. The unknown ray-dependent term $\log(ab/X)$ prevents a single global linear recombination with $\log(a/b)$ from producing $\log k$. Once the primitive ray is known, one can nevertheless subtract that term and divide by two. The GCD phase is therefore a direct ray-adapted observable, not a claim of more information than the complete ratio-product two-Mellin signal.

3 Dyadic factor-ray sectors and joint spacing

The two coordinates have different natural resolutions. To see them without mixing incompatible scales, let

$$\mathcal{D}_X := \{2^j : j \in \mathbb{Z}_{\geq 0}, 2^j \leq \beta X\} \quad (19)$$

and, for $R \in \mathcal{D}_X$, define the hyperbolic sector

$$\mathcal{J}_R(X) := \left\{ (a, b, k) : \begin{array}{l} (a, b) = 1, R \leq ab < 2R, \\ \alpha X \leq abk^2 \leq \beta X \end{array} \right\}. \quad (20)$$

Every coefficient supported by $W(abk^2/X)$ belongs to exactly one such sector. The cutoff $ak > D$ may remove indices but cannot worsen the spacing.

Lemma 3.1 (Explicit uniform lower gaps in one sector). *Let $(a, b, k), (c, d, \ell) \in \mathcal{J}_R(X)$. If $(a, b) \neq (c, d)$, then*

$$\left| \log \frac{a}{b} - \log \frac{c}{d} \right| \geq \delta_{\text{ray}}(R), \quad \delta_{\text{ray}}(R) := 2 \operatorname{arsinh} \left(\frac{1}{4R} \right). \quad (21)$$

If $k \neq \ell$, then

$$|\log k - \log \ell| \geq \delta_{\text{rad}}(R, X), \quad \delta_{\text{rad}}(R, X) := \log \left(1 + \sqrt{\frac{R}{\beta X}} \right). \quad (22)$$

Uniformly for $1 \leq R \leq \beta X$,

$$\delta_{\text{ray}}(R)^{-1} \asymp R, \quad \delta_{\text{rad}}(R, X)^{-1} \asymp_{\beta} \sqrt{X/R}. \quad (23)$$

Proof. For distinct reduced fractions, $|ad - bc| \geq 1$. The exact identity

$$2 \sinh \left(\frac{1}{2} \left| \log \frac{ad}{bc} \right| \right) = \frac{|ad - bc|}{\sqrt{abcd}}$$

and $ab, cd < 2R$ give

$$2 \sinh \left(\frac{1}{2} \left| \log \frac{a/b}{c/d} \right| \right) > \frac{1}{2R},$$

which implies (21).

Next, every supported radial index satisfies

$$k \leq \sqrt{\frac{\beta X}{ab}} \leq \sqrt{\frac{\beta X}{R}}.$$

If $1 \leq k < \ell$, then

$$\frac{\ell}{k} \geq 1 + \frac{1}{k} \geq 1 + \sqrt{\frac{R}{\beta X}},$$

and (22) follows. The final estimates use $\operatorname{arsinh} y \asymp y$ and $\log(1 + y) \asymp y$ for $0 < y \leq 1$, with fixed β absorbed into the constants. $\square$

Corollary 3.2 (The sectorwise resolution invariant). *The reciprocal gap scales obey*

$$\delta_{\text{ray}}(R)^{-1} \delta_{\text{rad}}(R, X)^{-2} \asymp_{\beta} X. \quad (24)$$

Thus the natural anisotropic resolution pair is

$$T_{\text{ray}} \asymp R, \quad T_{\text{rad}} \asymp \sqrt{X/R}, \quad T_{\text{ray}} T_{\text{rad}}^2 \asymp X. \quad (25)$$

Remark 3.3 (Separate inequalities, not a bandwidth trade). The last product is only a compact way to record the two gap scales. A very long radial window cannot distinguish two near-adjacent primitive rays having the same k , and a very long ray window cannot distinguish consecutive radial indices on one ray. Exact dephasing below therefore requires two separate threshold inequalities.

Remark 3.4 (Why the statement is sectorwise). Without coefficient-side sectorization, all $ab \leq \beta X$ occur. The worst primitive-ray gap is then of order X^{-1} , while the worst radial gap is of order $X^{-1/2}$. A single global rectangular band therefore needs

$$T_{\text{ray}} \asymp X, \quad T_{\text{rad}} \asymp \sqrt{X},$$

and has $T_{\text{ray}} T_{\text{rad}}^2 \asymp X^2$. One may instead filter each known dyadic sector at its own scales, but one may not replace that filter bank by one global band satisfying the smaller product law.

Remark 3.5 (Near-sharp populated interior families). The pairs of primitive rays $(N, 1)$ and $(N - 1, 1)$ have log gap $N^{-1} + O(N^{-2})$. On a fixed ray, consecutive radial indices K and $K - 1$ have log gap $K^{-1} + O(K^{-2})$. Choosing $N \asymp R$ and $K \asymp \sqrt{X/R}$ along populated families with $R \rightarrow \infty$ and $X/R \rightarrow \infty$ inside a nondegenerate hyperbolic shell shows that both scales in (23) are sharp up to constants for general supported coefficient systems. A particular empty or sparse sector may need only one of the two conditions, or neither.

4 Exact anisotropic dephasing

We use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-it\xi} dt$$

and the normalized Fejér kernel

$$\kappa(t) = \frac{2}{\pi} \left(\frac{\sin(t/2)}{t} \right)^2, \quad \widehat{\kappa}(\xi) = (1 - |\xi|)_+. \quad (26)$$

Here $\kappa(0) = 1/(2\pi)$ by continuity. In particular, $\int_{\mathbb{R}} \kappa = 1$, and the value of $\widehat{\kappa}$ at $|\xi| = 1$ is zero.

For arbitrary coefficients supported on $\mathcal{I}_R(X)$, write

$$\mathcal{O}_R(t, s) = \sum_{(a,b,k) \in \mathcal{I}_R(X)} x_{a,b,k} (a/b)^{it} k^{is}. \quad (27)$$

Proposition 4.1 (Exact pairwise Fejér criterion). *Let $T, S > 0$. For any finite set of distinct joint frequencies (λ_i, ρ_i) , tensor-Fejér coefficient recovery and the Parseval identity hold for every coefficient vector if and only if, for each $i \neq j$,*

$$T|\lambda_i - \lambda_j| \geq 1 \quad \text{or} \quad S|\rho_i - \rho_j| \geq 1. \quad (28)$$

Proof. After Fourier expansion, the multiplier of the (i, j) term is

$$\widehat{\kappa}(T(\lambda_i - \lambda_j)) \widehat{\kappa}(S(\rho_i - \rho_j)).$$

It vanishes exactly under (28). If the condition fails for one pair, both factors are strictly positive. A vector supported with equal phase on those two indices produces a nonzero cross term, and recovery of either coefficient has nonzero leakage. This proves necessity as well as sufficiency. $\square$

For a sparse fixed sector, (28) can be weaker than two coordinatewise gap conditions. The next theorem gives a uniform sufficient condition that depends only on R, X .

Theorem 4.2 (Exact sector recovery). *Suppose*

$$T\delta_{\text{ray}}(R) \geq 1, \quad S\delta_{\text{rad}}(R, X) \geq 1. \quad (29)$$

Then every $(a_0, b_0, k_0) \in \mathcal{I}_R(X)$ satisfies

$$x_{a_0, b_0, k_0} = \frac{1}{TS} \iint_{\mathbb{R}^2} \kappa(t/T) \kappa(s/S) \mathcal{O}_R(t, s) \times e^{-it \log(a_0/b_0) - is \log k_0} dt ds. \quad (30)$$

Moreover,

$$\iint_{\mathbb{R}^2} \kappa(t/T) \kappa(s/S) |\mathcal{O}_R(t, s)|^2 dt ds = TS \sum_{(a, b, k) \in \mathcal{I}_R(X)} |x_{a, b, k}|^2. \quad (31)$$

Proof. Scaling (26) gives

$$\int_{\mathbb{R}} \kappa(t/T) e^{it(\lambda - \lambda_0)} dt = T \hat{\kappa}(T(\lambda_0 - \lambda)). \quad (32)$$

Insert (27) into (30). If the primitive rays differ, the first Fourier factor vanishes by (21)–(29). If the rays agree but the triples differ, then their radial indices differ and the second factor vanishes by (22). Only the selected joint frequency remains. This proves coefficient recovery.

Expanding the square in (31) gives the same two Fourier factors for every pair of triples. Every off-diagonal pair is killed in at least one coordinate, while a diagonal pair contributes $TS|x_{a, b, k}|^2$. $\square$

The theorem is exact at the gap thresholds because the compact Fourier transform of the kernel vanishes at its endpoints. It uses infinite physical tails in (t, s) ; a hard rectangle behaves differently.

Theorem 4.3 (Hard-rectangle mean square). *Let $I, J \subset \mathbb{R}$ be bounded intervals of lengths L and M , respectively. For coefficients supported on $\mathcal{I}_R(X)$,*

$$\iint_{I \times J} |\mathcal{O}_R(t, s)|^2 dt ds = LM \sum |x_{a, b, k}|^2 + O\left(\left\{\frac{M}{\delta_{\text{ray}}} + \frac{L}{\delta_{\text{rad}}} + \frac{1}{\delta_{\text{ray}}\delta_{\text{rad}}}\right\} \sum |x_{a, b, k}|^2\right), \quad (33)$$

where the gaps are those in lemma 3.1 and the implied constant is absolute. Consequently (33) is an asymptotic uniformly for arbitrary coefficients if

$$L\delta_{\text{ray}} \longrightarrow \infty, \quad M\delta_{\text{rad}} \longrightarrow \infty. \quad (34)$$

Proof. Group first by primitive ray:

$$\mathcal{O}_R(t, s) = \sum_{(a, b)} F_{a, b}(s) e^{it \log(a/b)}, \quad F_{a, b}(s) = \sum_k x_{a, b, k} e^{is \log k}.$$

For every fixed s , the generalized Hilbert inequality of Montgomery and Vaughan [7] and the ray gap give

$$\int_I |\mathcal{O}_R(t, s)|^2 dt = L \sum_{(a, b)} |F_{a, b}(s)|^2 + O\left(\delta_{\text{ray}}^{-1} \sum_{(a, b)} |F_{a, b}(s)|^2\right).$$

Integrate in s . A second application of the same inequality, now to the δ_{rad} -separated frequencies $\log k$ on each ray, gives

$$\int_J |F_{a, b}(s)|^2 ds = M \sum_k |x_{a, b, k}|^2 + O\left(\delta_{\text{rad}}^{-1} \sum_k |x_{a, b, k}|^2\right).$$

Substitution and expansion of the two error factors proves (33). Division by $LM \sum |x|^2$ gives the last assertion. $\square$

At the critical scales $L \asymp R$ and $M \asymp \sqrt{X/R}$, the hard-rectangle error is only bounded relative to the main term. Exact equality at that scale is a special advantage of the compact-Fourier Fejér kernel.

4.1 Global and multiresolution forms

For the undecomposed signal, put

$$\delta_X := 2 \operatorname{arsinh}\left(\frac{1}{2\beta X}\right), \quad \Delta_X := \log\left(1 + \frac{1}{\sqrt{\beta X}}\right). \quad (35)$$

The first is the global hyperbolic ratio gap from TPC-11; the second follows from $k \leq \sqrt{\beta X}$. The same proof, using the global gaps in (35), gives the following deliberately more expensive statement.

Corollary 4.4 (Global exact radial Parseval identity). *If $T\delta_X \geq 1$ and $S\Delta_X \geq 1$, then*

$$\iint_{\mathbb{R}^2} \kappa(t/T) \kappa(s/S) |\mathcal{R}_{h,D}^{\operatorname{rad}}(X; t, s)|^2 dt ds = TS \mathcal{F}_{h,D}(X). \quad (36)$$

The analogous identity holds for the raw transform and its fully diagonal energy.

There is also an exact sector-adapted filter bank. Let $\mathcal{R}_{h,D;R}^{\operatorname{rad}}$ denote the restriction of (5) to $R \leq ab < 2R$. Choose arbitrary T_R, S_R satisfying (29) in each nonempty sector. Then

$$\sum_{\substack{R \in \mathcal{D}_X \\ \mathcal{I}_R(X) \neq \emptyset}} \frac{1}{T_R S_R} \iint_{\mathbb{R}^2} \kappa(t/T_R) \kappa(s/S_R) |\mathcal{R}_{h,D;R}^{\operatorname{rad}}(X; t, s)|^2 dt ds = \mathcal{F}_{h,D}(X). \quad (37)$$

This is the rigorous meaning of using the sectorwise law $T_R S_R^2 \asymp X$: the sectors are isolated before their squared observations are combined.

Remark 4.5 (Geometric sharpness along interior families only). Two equal- k coefficients on the near-adjacent rays $(N, 1)$ and $(N - 1, 1)$ retain a nonzero Fejér cross term if $T = o(N)$. Two coefficients at consecutive k on one ray retain a nonzero cross term if $S = o(k)$. Along populated interior families containing these configurations, neither separate threshold can be uniformly reduced for arbitrary coefficient systems. A sparse fixed sector can require less. This is a conditioning and resolution statement; it does not assert that the actual arithmetic coefficients saturate every geometric obstruction.

5 Arithmetic ledgers and short-interval transfer

The geometric lift changes the spectral fibres but not the product centers to which the short-interval theorem is applied. For $r \geq 1$ put

$$A_D(r) := |W(r/X)|^2 \sum_{\substack{n|r \\ n > D}} u(n)^2, \quad (38)$$

$$B_{D,t,s}^{\operatorname{rad}}(r) := W(r/X) \sum_{\substack{n|r \\ n > D}} u(n) (n^2/r)^{it} \gcd(n, r/n)^{is}. \quad (39)$$

The first weight is unchanged from TPC-12. The second is the coefficient obtained by grouping (4) by $r = mn$.

Lemma 5.1 (Product ledgers). *Uniformly in $D \geq 1$ and $t, s \in \mathbb{R}$,*

$$A_D(r) \ll_W \mathbf{1}_{[\alpha X, \beta X]}(r) ((\log(2r))^2 + \tau(r)), \quad (40)$$

$$|B_{D,t,s}^{\text{rad}}(r)| \ll_W \mathbf{1}_{[\alpha X, \beta X]}(r) (\log(2r) + \tau(r)), \quad (41)$$

and

$$\sum_r A_D(r)^2 \ll_W X (\log(2X))^4, \quad (42)$$

$$\sum_r |B_{D,t,s}^{\text{rad}}(r)|^2 \ll_W X (\log(2X))^3. \quad (43)$$

Proof. Since $(\Lambda - 1)^2 \leq 2(1 + \Lambda^2)$,

$$\sum_{d|r} u(d)^2 \leq 2\tau(r) + 2 \sum_{p^j|r} (\log p)^2 \leq 2\tau(r) + 2(\log r)^2.$$

Also, both phases in (39) have modulus one, so

$$|B_{D,t,s}^{\text{rad}}(r)| \leq \|W\|_\infty \sum_{d|r} |u(d)| \leq \|W\|_\infty (\log r + \tau(r)).$$

The second moments follow from $\sum_{r \leq y} \tau(r)^2 \ll y(\log(2y))^3$. This is the same proof as the product ledger in TPC-12; the new GCD phase costs nothing. $\square$

Define the total source diagonal

$$\mathcal{S}_D(X) := \sum_r A_D(r) = \sum_{\substack{n \leq D \\ m \geq 1}} u(n)^2 |W(mn/X)|^2. \quad (44)$$

Lemma 5.2 (Source diagonal). *For every fixed $\eta > 0$, uniformly for $1 \leq D \leq X^{1-\eta}$,*

$$\mathcal{S}_D(X) = \frac{1}{2} X I_2(W) ((\log X)^2 - (\log D)^2) + O_W(X \log(2X)). \quad (45)$$

Proof. Euler summation gives

$$\sum_{m \geq 1} |W(mn/X)|^2 = \frac{X}{n} I_2(W) + O_W(1).$$

The prime number theorem and partial summation give

$$\sum_{n \leq y} \frac{(\Lambda(n) - 1)^2}{n} = \frac{1}{2} (\log y)^2 - \log y + O(1),$$

while $\sum_{n \leq y} u(n)^2 \ll y \log(2y)$. Subtracting the range $n \leq D$ proves the assertion; see also Iwaniec and Kowalski [6, Chapter 5] and TPC-12 [11]. $\square$

The following transferred moment is the deep arithmetic input. It is quoted in the form proved as Proposition 4.2 of TPC-12. Its proof starts from Corollary 1.4 of arXiv:2405.20552v2, the Guth–Maynard prime number theorem for almost all intervals of length $X^{2/15+\varepsilon}$ [3]. The exceptional set has size $O(X \exp\{-c(\log X)^{1/4}\})$ for each discretized interval length; TPC-12 takes a controlled union over the lengths needed to smooth G . Cauchy–Schwarz and (42) then show that the exceptional centers carry negligible A_D -mass. For $H \leq X^{0.98}$ this is the almost-all argument. For $H > X^{0.98}$, TPC-12 partitions the full shift interval into equal blocks of lengths in $[X^{0.97}, 2X^{0.97}]$ and applies the uniform 17/30 short-interval theorem on every block; there is no exceptional center in that range. These two cases cover the full upper range $H \leq X$ stated below.

Proposition 5.3 (Weighted target-square transfer). *Let $G \in C_c^1((0, \infty))$ be nonnegative and nonzero. Fix $0 < \varepsilon < 13/15$ and $0 < \eta < 1$. Uniformly for (9),*

$$\begin{aligned} \sum_r A_D(r) \sum_h G(h/H) (\Lambda(r+h) - 1)^2 \\ = HI_0(G) \log X \mathcal{S}_D(X) + O_{\varepsilon, W, G}(HX(\log X)^2). \end{aligned} \quad (46)$$

The same formula holds with the target square replaced by $\Lambda(r+h)^2$.

Proof. This is Proposition 4.2 of Wang [11]. Outside the transferred exceptional set, smooth partial summation gives

$$\sum_h G(h/H) (\Lambda(r+h) - 1)^2 = HI_0(G) \log X + O(H).$$

The $O(H)$ term sums to $O(H\mathcal{S}_D) = O(HX(\log X)^2)$. On the exceptional set, the pointwise $O(H(\log X)^2)$ bound and the arbitrarily large logarithmic saving in its A_D -mass give a smaller contribution. The raw square is identical. $\square$

The signed transfer also survives the new phase.

Theorem 5.4 (Radially completed short-shift first moment). *Under (9), for every $A > 0$, uniformly in $t, s \in \mathbb{R}$,*

$$\frac{1}{H} \sum_h G(h/H) \mathcal{T}_{h,D}^{\text{rad}}(X; t, s) = I_0(G) \mathcal{M}_D^{\text{rad}}(X; t, s) + O_{A, \varepsilon, W, G}(X(\log X)^{-A}). \quad (47)$$

Proof. Grouping by $r = mn$ gives

$$\frac{1}{H} \sum_h G(h/H) \mathcal{R}_{h,D}^{\text{rad}}(X; t, s) = \sum_r B_{D,t,s}^{\text{rad}}(r) \left\{ \frac{1}{H} \sum_h G(h/H) (\Lambda(r+h) - 1)^2 \right\}.$$

On regular centers the braces save every fixed power of $\log X$. Equation (41) and the mean order of τ give $\sum_r |B_{D,t,s}^{\text{rad}}(r)| \ll_W X \log(2X)$. On the exceptional set, Cauchy–Schwarz, (43), and its stretched-exponential cardinality again save every logarithmic power. All bounds are uniform in both frequency variables because the added phase has modulus one. Finally the Riemann-sum normalization of G produces the displayed model term. $\square$

6 Short-shift radial energy and spectral laws

The benefit of the added coordinate appears before any estimate is used.

Proposition 6.1 (Exact product-center identity). *For every h, D, X ,*

$$\mathcal{F}_{h,D}(X) = \sum_r A_D(r) (\Lambda(r+h) - 1)^2. \quad (48)$$

If

$$\mathcal{F}_{h,D}^{\text{raw}}(X) := \sum_{\substack{n > D \\ m \geq 1}} u(n)^2 \Lambda(mn+h)^2 |W(mn/X)|^2, \quad (49)$$

then

$$\mathcal{F}_{h,D}^{\text{raw}}(X) = \sum_r A_D(r) \Lambda(r+h)^2. \quad (50)$$

Proof. By [proposition 2.1](#), summing over primitive triples is the same as summing once over every factor pair (m, n) . Group those pairs by $r = mn$. The source squares on one product center sum to $A_D(r)$, proving both identities. $\square$

No target with a different radial index occurs in (48). In particular, no two-target upper sieve is needed.

Theorem 6.2 (Short-shift fully radial energy). *Let $G \in C_c^1((0, \infty))$ be nonnegative and nonzero. Fix $0 < \varepsilon < 13/15$ and $0 < \eta < 1$. Uniformly for (9),*

$$\begin{aligned} \sum_h G(h/H) \mathcal{F}_{h,D}(X) &= \frac{1}{2} HX I_0(G) I_2(W) \log X \\ &\quad \times ((\log X)^2 - (\log D)^2) + O_{\varepsilon, \eta, W, G}(HX(\log X)^2). \end{aligned} \quad (51)$$

The same asymptotic holds for $\mathcal{F}_{h,D}^{\text{raw}}(X)$.

Proof. Insert (48) and apply [proposition 5.3](#). This gives

$$\sum_h G(h/H) \mathcal{F}_{h,D}(X) = HI_0(G) \log X \mathcal{S}_D(X) + O(HX(\log X)^2).$$

Now substitute (45). Its remainder, after multiplication by $H \log X$, is also $O(HX(\log X)^2)$. The raw result follows from (50) and the raw part of [proposition 5.3](#). $\square$

If $W \neq 0$, then $I_2(W) > 0$ and

$$(\log X)^2 - (\log D)^2 \geq (2\eta - \eta^2)(\log X)^2.$$

Thus the relative error in (51) is $O_\eta(1/\log X)$.

Remark 6.3 (A changed observable, not a stronger estimate of the old one). The ratio-only ray energy of TPC-12 is

$$\mathcal{E}_{h,D}(X) = \sum_{(a,b)=1} \left| \sum_k x_{a,b,k} \right|^2 = \mathcal{F}_{h,D}(X) + \sum_{(a,b)=1} \sum_{k \neq \ell} x_{a,b,k} \overline{x_{a,b,\ell}}. \quad (52)$$

The $O(HX(\log X)^2 \log \log X)$ error in TPC-12 bounds the last sum after shift averaging by a two-linear-forms Selberg upper sieve. The present $O(HX(\log X)^2)$ error is smaller because (7) defines a different, fully separated observable in which the off-diagonal is absent. We do not claim an improved error for $\mathcal{E}_{h,D}$ itself.

6.1 Global two-dimensional Fejér band

Combining the arithmetic theorem with [corollary 4.4](#) gives a single-band statement.

Corollary 6.4 (Short-shift global joint band). *If $T\delta_X \geq 1$ and $S\Delta_X \geq 1$, then under the hypotheses of [theorem 6.2](#),*

$$\begin{aligned} \sum_h G(h/H) \iint_{\mathbb{R}^2} \kappa(t/T) \kappa(s/S) |\mathcal{R}_{h,D}^{\text{rad}}(X; t, s)|^2 dt ds \\ = \frac{1}{2} TSHX I_0(G) I_2(W) \log X ((\log X)^2 - (\log D)^2) \\ + O_{\varepsilon, \eta, W, G}(TSHX(\log X)^2). \end{aligned} \quad (53)$$

The same statement holds for the raw transform.

Proof. For every h , (36) turns the integral into $T\mathcal{S}\mathcal{F}_{h,D}(X)$. Sum [theorem 6.2](#). $\square$

6.2 Sector-adapted multiresolution band

For each nonempty R -sector, choose T_R, S_R satisfying (29). Define

$$\mathfrak{P}_{h,D}(X) := \sum_{\substack{R \in \mathcal{D}_X \\ \mathcal{I}_R(X) \neq \emptyset}} \frac{1}{T_R S_R} \iint_{\mathbb{R}^2} \kappa(t/T_R) \kappa(s/S_R) \times |\mathcal{R}_{h,D;R}^{\text{rad}}(X; t, s)|^2 dt ds. \quad (54)$$

Equation (37) says exactly that

$$\mathfrak{P}_{h,D}(X) = \mathcal{F}_{h,D}(X). \quad (55)$$

We therefore obtain the sector-optimal version without needing an individual arithmetic asymptotic for each R .

Corollary 6.5 (Short-shift multiresolution spectrum). *Under (9),*

$$\begin{aligned} \sum_h G(h/H) \mathfrak{P}_{h,D}(X) &= \frac{1}{2} H X I_0(G) I_2(W) \log X \\ &\times ((\log X)^2 - (\log D)^2) + O(H X (\log X)^2). \end{aligned} \quad (56)$$

This corollary combines sectorwise spectral resolutions only after each sector has been isolated and normalized. It neither asserts a common small global band nor evaluates a specified fixed shift.

7 Parity-conditioned even shifts

Restricting the ensemble to even shifts requires its own local centering. For $h \equiv 0 \pmod{2}$ and a product center r , put

$$\eta_r^{(2)}(h) := \Lambda(r + h) - 2\mathbf{1}_{2 \nmid r}. \quad (57)$$

Define

$$x_{a,b,k}^{(2)}(h; X, D) := u(ak) \eta_{abk^2}^{(2)}(h) W(abk^2/X) \quad (58)$$

and the fully radial parity-centered energy

$$\mathcal{F}_{h,D}^{(2)}(X) := \sum_{(a,b)=1} \sum_{\substack{k \geq 1 \\ ak > D}} |x_{a,b,k}^{(2)}(h; X, D)|^2. \quad (59)$$

The associated joint signal is obtained by multiplying $x_{a,b,k}^{(2)}$ by $(a/b)^{it} k^{is}$.

Theorem 7.1 (Short even-shift radial energy). *Let $G \in C_c^1((0, \infty))$ be nonnegative and nonzero. Under (9),*

$$\begin{aligned} \sum_{h \equiv 0 \pmod{2}} G(h/H) \mathcal{F}_{h,D}^{(2)}(X) &= \frac{1}{4} H X I_0(G) I_2(W) \log X \\ &\times ((\log X)^2 - (\log D)^2) + O_{\varepsilon, \eta, W, G}(H X (\log X)^2). \end{aligned} \quad (60)$$

Proof. The coordinate bijection gives

$$\mathcal{F}_{h,D}^{(2)}(X) = \sum_r A_D(r) |\eta_r^{(2)}(h)|^2.$$

For odd r , the transferred target-square moment over even shifts is

$$\sum_{h \equiv 0 \pmod{2}} G(h/H) |\eta_r^{(2)}(h)|^2 = H I_0(G) \log X + O(H)$$

outside the same stretched-exponentially sparse set used in TPC-12. For even r , the target term is supported on powers of two. Since $r + h \asymp X$ on the shift support, only $O_G(1)$ such powers occur and

$$\sum_{h \equiv 0 \pmod{2}} G(h/H) |\eta_r^{(2)}(h)|^2 \ll_G (\log X)^2.$$

After multiplication by $A_D(r)$ and summation, these rows contribute

$$O_W(X(\log X)^4) = o(HX(\log X)^2)$$

in the stated range. The exceptional-set transfer applies to the odd rows with the same A_D ledger, and their $O(H)$ errors sum to $O(H\mathcal{S}_D)$.

It remains to sum the source squares on odd product centers. TPC-12 proves

$$\sum_{\substack{n > D, m \geq 1 \\ 2 \nmid nm}} u(n)^2 |W(mn/X)|^2 = \frac{1}{4} X I_2(W) ((\log X)^2 - (\log D)^2) + O_W(X \log X). \quad (61)$$

The factor $1/2$ from odd m multiplies the $1/2$ in the full source diagonal. Restricting n to odd values changes only the lower-order linear logarithm. Multiplication by $H \log X$ and the transferred exceptional error give (60). $\square$

The normalized conditional average uses $2/H$ rather than $1/H$ because only half of the shifts are sampled. Accordingly, after conditional normalization, the leading source coefficient is $1/2$; the unnormalized sum in (60) has coefficient $1/4$.

All exact dephasing statements in [section 4](#) apply coefficient by coefficient to $x_{a,b,k}^{(2)}$. For example, the even-shift version of the global joint band has the right-hand side of (53) with $1/2$ replaced by $1/4$ and the same $O(TSHX(\log X)^2)$ error.

Remark 7.2 (No selected even gap). The support of $G(h/H)$ moves to infinity with H . The theorem conditions the growing ensemble on parity but contains no prescribed even integer once X is large. In particular, it gives no information about the single shift $h = 2$.

8 The fixed-shift arithmetic boundary

The completed signal can recover every finite coefficient, but recovery is not evaluation. Two exact slices locate the remaining arithmetic difficulty.

8.1 The endpoint prime-pair slice

The factor endpoint $m = 1$ is characterized in primitive coordinates by

$$(a, b, k) = (n, 1, 1).$$

Define its fully diagonal energy, independently of the source cutoff, by

$$\mathcal{F}_h^{\text{end}}(X) := \sum_{n \geq 1} (\Lambda(n) - 1)^2 (\Lambda(n+h) - 1)^2 |W(n/X)|^2. \quad (62)$$

For $D \leq X^{1-\eta}$ this is exactly the $m = 1$ slice of $\mathcal{F}_{h,D}(X)$ once X is large, because $n \geq \alpha X > D$ on the support.

Proposition 8.1 (Endpoint reduction). *For every fixed nonzero integer h ,*

$$\mathcal{F}_h^{\text{end}}(X) = (\log X)^2 \sum_n \Lambda(n) \Lambda(n+h) |W(n/X)|^2 + O_{h,W}(X \log X). \quad (63)$$

Proof. On the support, $n, n+h \asymp_W X$. If both are prime, then

$$(\log n - 1)^2 (\log(n+h) - 1)^2 = (\log X)^2 \log n \log(n+h) + O_W((\log X)^3).$$

The Selberg upper-bound sieve gives $O_{h,W}(X/(\log X)^2)$ simultaneous prime values, so their total error is $O(X \log X)$. If exactly one of the two integers is prime and the other is not a prime power, the summand on the left is $O((\log X)^2)$; the prime number theorem bounds the total by $O(X \log X)$. If neither is a prime power, the total is $O(X)$.

Finally, the number of proper prime powers in an interval of size $O(X)$ is $O(\sqrt{X})$ up to a harmless logarithmic factor. Giving all terms involving one such value the trivial weight $O((\log X)^4)$ produces $o(X \log X)$. These cases also cover the proper-prime-power part of the von Mangoldt correlation on the right. $\square$

For a fixed nonzero shift, let

$$\mathfrak{S}(h) := \begin{cases} 2C_2 \prod_{\substack{p|h \\ p>2}} \frac{p-1}{p-2}, & 2 \mid h, \\ 0, & 2 \nmid h, \end{cases} \quad C_2 := \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right). \quad (64)$$

This is the Hardy–Littlewood prime-pair singular series [5]; it is not assumed in the positive theorems.

Corollary 8.2 (Exact Hardy–Littlewood strength threshold). *For every fixed even $h \neq 0$, the asymptotic*

$$\mathcal{F}_h^{\text{end}}(X) = \mathfrak{S}(h) I_2(W) X (\log X)^2 + o_{h,W}(X (\log X)^2) \quad (65)$$

holds if and only if

$$\sum_n \Lambda(n) \Lambda(n+h) |W(n/X)|^2 = \mathfrak{S}(h) I_2(W) X + o_{h,W}(X). \quad (66)$$

For $h = 2$, this is the corresponding weighted twin-prime Hardy–Littlewood asymptotic.

Proof. Subtract (63) and divide by $(\log X)^2$. Its error becomes $O(X/\log X) = o(X)$, proving both implications. $\square$

When $W \neq 0$, (66) at $h = 2$ would in particular imply infinitely many twin primes after proper prime powers are removed. Neither side of the equivalence is proved here. For odd fixed h , elementary parity and the same case split as above give only the much smaller statement

$$\mathcal{F}_h^{\text{end}}(X) = O_{h,W}(X \log X). \quad (67)$$

Indeed, a fixed odd difference excludes two prime values at height X ; proper prime powers are negligible, and the terms with exactly one prime have total size $O(X \log X)$.

8.2 A fixed quadratic ray persists

There is an earlier obstruction before the endpoint correlation. On the primitive ray $(a, b) = (6, 1)$ and at $h = 1$, the source value satisfies $u(6k) = -1$ for every $k \geq 1$. Define the cutoff-free comparison

$$Q_6(X) := \sum_{k \geq 1} (\Lambda(6k^2 + 1) - 1)^2 |W(6k^2/X)|^2. \quad (68)$$

If $D < \sqrt{6\alpha X}$, then every supported k satisfies $6k > D$, so $Q_6(X)$ is exactly the $(6, 1)$ -ray slice of $\mathcal{F}_{1,D}(X)$. For larger D we make no such identification. Put

$$P_6(X) := \sum_{k \geq 1} \Lambda(6k^2 + 1) |W(6k^2/X)|^2. \quad (69)$$

Proposition 8.3 (Quadratic-prime strength threshold). *One has*

$$Q_6(X) = \log X P_6(X) + O_W(\sqrt{X}). \quad (70)$$

Consequently, for every fixed constant C ,

$$Q_6(X) = C\sqrt{X} \log X + o(\sqrt{X} \log X) \quad (71)$$

if and only if

$$P_6(X) = C\sqrt{X} + o(\sqrt{X}). \quad (72)$$

Proof. Put $K \asymp_W \sqrt{X}$ for the supported k -range. The dimension-one Selberg upper-bound sieve for the irreducible quadratic $6k^2 + 1$ gives

$$\#\{k \asymp_W K : 6k^2 + 1 \text{ is prime}\} \ll_W \frac{K}{\log K};$$

see Halberstam and Richert [4]. Restoring the logarithmic weight yields $P_6(X) \ll_W K \ll_W \sqrt{X}$.

For $r = 1, 2$, the proper-prime-power contribution satisfies

$$\sum_{\substack{k \asymp_W \sqrt{X} \\ 6k^2+1=p^j, j \geq 2}} \Lambda(6k^2 + 1)^r |W(6k^2/X)|^2 \ll_W X^{1/3} (\log X)^{r+1} + (\log X)^{r+1} = o_W(\sqrt{X}).$$

For even j this follows from the associated Pell equation, which has $O_W(\log X)$ relevant solutions; for odd $j \geq 3$, there are $O_W(X^{1/3} \log X)$ candidates. These estimates are detailed in TPC-12 [11].

At prime values in the support, $\log(6k^2 + 1) = \log X + O_W(1)$. Therefore

$$\begin{aligned} Q_6(X) &= \sum_k \Lambda(6k^2 + 1)^2 |W(6k^2/X)|^2 - 2P_6(X) + \sum_k |W(6k^2/X)|^2 \\ &= \log X P_6(X) + O_W(P_6(X)) + O_W(\sqrt{X}) = \log X P_6(X) + O_W(\sqrt{X}). \end{aligned}$$

The final equivalence follows after division by $\log X$. $\square$

The asymptotic (72) is a weighted prime number theorem for the fixed polynomial $6k^2 + 1$ and is open. The radial coordinate separates the individual k -labels but supplies no new distribution theorem for their prime targets.

8.3 What is lost after retaining only total energy

The full two-coordinate signal is injective, but the scalar $\mathcal{F}_{h,D} = \sum |x_{a,b,k}|^2$ again forgets where its energy lies. The following coefficient-blind minimax identity, for arbitrary coefficient vectors, prevents an endpoint conclusion from the total energy alone.

Proposition 8.4 (Exact energy-only minimax boundary). *Let a finite index set I be partitioned into nonempty sets J and $I \setminus J$. For $x \in \mathbb{C}^I$, define*

$$S(x) = \sum_{i \in I} |x_i|^2, \quad Q_J(x) = \sum_{i \in J} |x_i|^2.$$

Then, for every $B > 0$,

$$\inf_{\mathcal{A}: [0, B^2] \rightarrow \mathbb{R}} \sup_{\|x\|_2 \leq B} |Q_J(x) - \mathcal{A}(S(x))| = \frac{B^2}{2}. \quad (73)$$

Proof. Take two vectors of norm B , one supported in J and one in its complement. They have the same observed value $S = B^2$, while their targets are B^2 and 0. Every estimator therefore has worst-case error at least $B^2/2$. Conversely, the estimator $\mathcal{A}(S) = S/2$ obeys $|Q_J - S/2| \leq S/2 \leq B^2/2$. $\square$

For the endpoint, J is the set $b = k = 1$. The complete joint signal can recover these coefficients by (30), but the total-energy asymptotic (51) cannot separate them from the bulk. In particular, $\mathcal{F}_{h,D} \geq \mathcal{F}_h^{\text{end}}$ supplies no twin-prime lower bound: the bulk may account for the entire known main term. For $W \neq 0$ and large X , both the endpoint and its complement are nonempty in the stated source-cutoff range. The minimax statement does not rule out an estimator using additional special arithmetic information about the von Mangoldt coefficients.

The classical sieve parity limitation explains why upper-bound sieve technology alone does not turn these reductions into lower bounds [2, 8]. The present paper repairs a geometric label collision and then stops exactly where new arithmetic input is required.

9 Finite certificate and reproducibility

The analytic asymptotic imports the Guth–Maynard short-interval theorem and the weighted transfer proved in TPC-12. No finite computation can verify those asymptotic inputs. The included deterministic certificate checks the exact new algebra and the resolution interfaces.

The script `experiments/radial_mellin_certificate.py` performs the following checks.

- (i) It enumerates a finite factor box and verifies the bijection $(m, n) \leftrightarrow (a, b, k)$, including reconstruction and injectivity of $(a/b, k)$.
- (ii) It compares the direct factor-pair transform with the compressed two-coordinate expansion at several pairs (t, s) .
- (iii) In every dyadic sector containing at least two distinct rays or radial indices, respectively, it computes the corresponding actual minimum log gap and verifies the lower bounds (21)–(22).
- (iv) It evaluates the tensor Fourier-side Fejér quadratic form at admissible scale parameters and verifies that every off-diagonal multiplier is zero, leaving exactly $TS \sum |x_{a,b,k}|^2$. It also checks the coefficient-recovery multipliers.
- (v) It groups the same squared coefficients by $r = mn$ and verifies the exact identity $\mathcal{F}_{h,D} = \sum_r A_D(r)(\Lambda(r+h) - 1)^2$.
- (vi) It constructs the two extremal vectors in [proposition 8.4](#), confirming that identical total energy can place all mass either in the endpoint or in its complement.

The companion unit tests repeat these identities at independent parameter values. Integer factorization and GCD data are exact; floating-point tolerances occur only in logarithms and phases. The commands are

```
python experiments/radial_mellin_certificate.py
python -m unittest experiments.test_radial_mellin_certificate
```

The saved JSON output is a regression certificate for finite identities, not numerical evidence for a fixed-shift prime-pair or quadratic-prime asymptotic.

10 Conclusion and next analytic interface

The coordinate omitted from the ratio-only factor-ray spectrum can be restored exactly after the observable is augmented with GCD data; it is not recovered from ratio-only samples. The GCD–Mellin lift replaces each collapsed radial fibre by the injective joint frequency

$$(\log(a/b), \log k).$$

On $ab \asymp R$, its two resolution scales are R and $\sqrt{X/R}$; tensor Fejér kernels recover individual coefficients and their energy exactly. A tensor Montgomery–Vaughan argument supplies a stable hard-rectangle asymptotic once both side lengths exceed their reciprocal-gap scales by unbounded factors. This constructs an exact two-parameter diagonal unitary representation of the augmented coefficient array; no intrinsic prime, logistic, or transfer-operator spectrum is claimed.

The arithmetic effect is equally explicit. Full radial dephasing changes the ray energy into the pure product-center target-square diagonal. TPC-12’s weighted exceptional-set transfer then yields the short-shift asymptotic with error $O(HX \log^2 X)$, removing the $\log \log X$ cost of the same-ray two-target upper sieve. The gain is real, but it belongs to a new observable; the old ratio-only energy still contains its coherent radial cross terms.

There is no corresponding fixed-shift breakthrough. Exact inversion reveals the arithmetic coefficients rather than estimating them. The $m = 1$ slice is equivalent at natural precision to a weighted Hardy–Littlewood correlation, and the cutoff-free $(6, 1)$ ray comparison is equivalent at natural precision to a weighted prime number theorem for $6k^2 + 1$. Retaining only total radial energy loses the endpoint label again, with the exact minimax loss in (73). These are positive and negative statements at matching normalizations.

The next honest interface is therefore not a smaller formal bandwidth or the substitution $H = 1$. It is an arithmetic, label-preserving signed observable that can distinguish an endpoint or a controlled family of radial slices before squaring, together with an explicit source of Type-II or dispersion cancellation not reducible to an upper-bound sieve. A viable next theorem may average an additional arithmetic parameter while retaining a specified radial selector. A negative result would be equally informative if it proves that every proposed average still factors through parity-blind local data. In either case, the endpoint and quadratic-ray equivalences above are mandatory tests.

No fixed-shift prime-pair asymptotic, twin-prime lower bound, quadratic-prime theorem, new distribution level, or violation of sieve parity has been obtained.

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